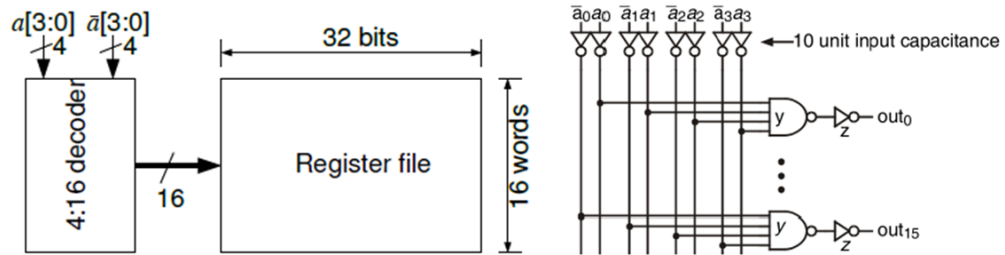


Tutorial 8

1. A decoder has to be designed for a register file in the embedded processor for automotive applications. The block level diagram is given below. The decoder has following



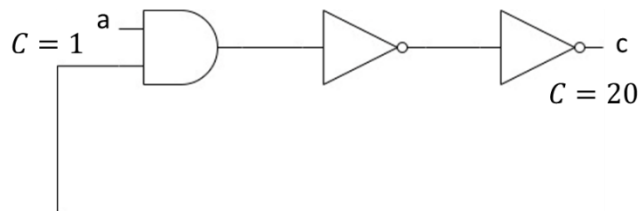
specifications.

- a. 16 word register file.
- b. Each word is 32 bits wide.
- c. Each bit presents a load of 3 unit-sized transistors.
- d. True and complementary inputs of address bits $a[3:0]$ are available.
- e. Each input may drive 10 unit-sized transistors.

The designer needs to decide:

- (a) How many stages to use?
- (b) How large should each gate be?
- (c) How fast can the decoder operate? (Approximate results to lower and upper integers for number of stages and do calculations for both cases).

2. A logic path has been designed as follows. All gates are LO-skew gates.



- a. Size the path for min average delay. What is the minimum and average delay.
 - b. Size the path for minimizing the propagation delay for a rising transition at the input
3. Consider a pseudo-NMOS inverter fabricated in 0.25um CMOS technology for which $k'_n = 115 \mu A/V^2$, $k'_p = 30 \mu A/V^2$, $V_{Tn} = -V_{Tp} = 0.5 V$ and $V_{DD} = 2.5 V$. The size of the PMOS is 0.25um/0.25um and the NMOS is chosen such that $k_n = 9k_p$. Find:
 - a. $V_{OH}, V_{OL}, V_{IH}, V_{IL}, NM_L, NM_H$
 - b. The static leakage current and the power dissipation
 - c. Assuming a load capacitance of 7fF, t_{pHL} , t_{pLH} and t_p .

4. In order to work correctly, what should be the setup and the hold time at input A in the circuit. Also find the maximum operating frequency of the circuit.

